

GDU Outlook

NC 09-90 PCE

Missouri Valley, IA 51555 · planted May 1, 2026



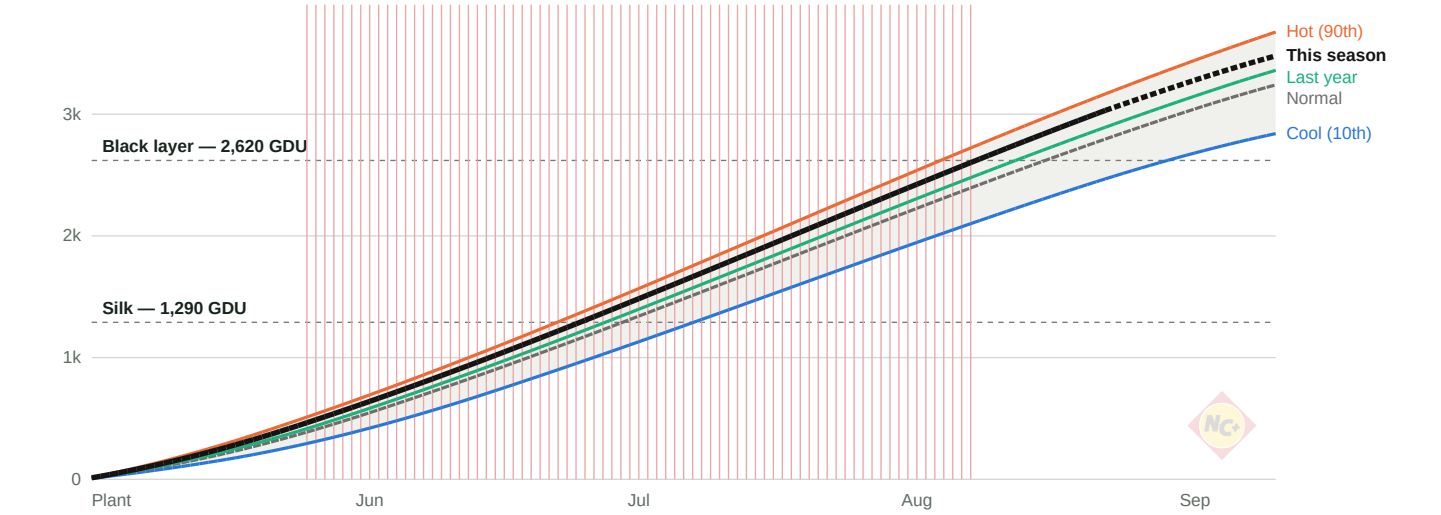
The Hybrid

Relative maturity	109 day
GDUs to silk	1,290
GDUs to black layer	2,620

2,602 GDU ACCUMULATED	+208 GDU AHEAD OF NORMAL	99 DAYS SINCE PLANTING
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Observed through Aug 7, 2026, then the 16-day forecast, then projected.

GDU Accumulation



Vertical rules: red = the 75 days the high hit 86 °F, where heat above the cap added nothing.
Solid = observed through Aug 7 plus forecast through Aug 22. Dashed = projected.

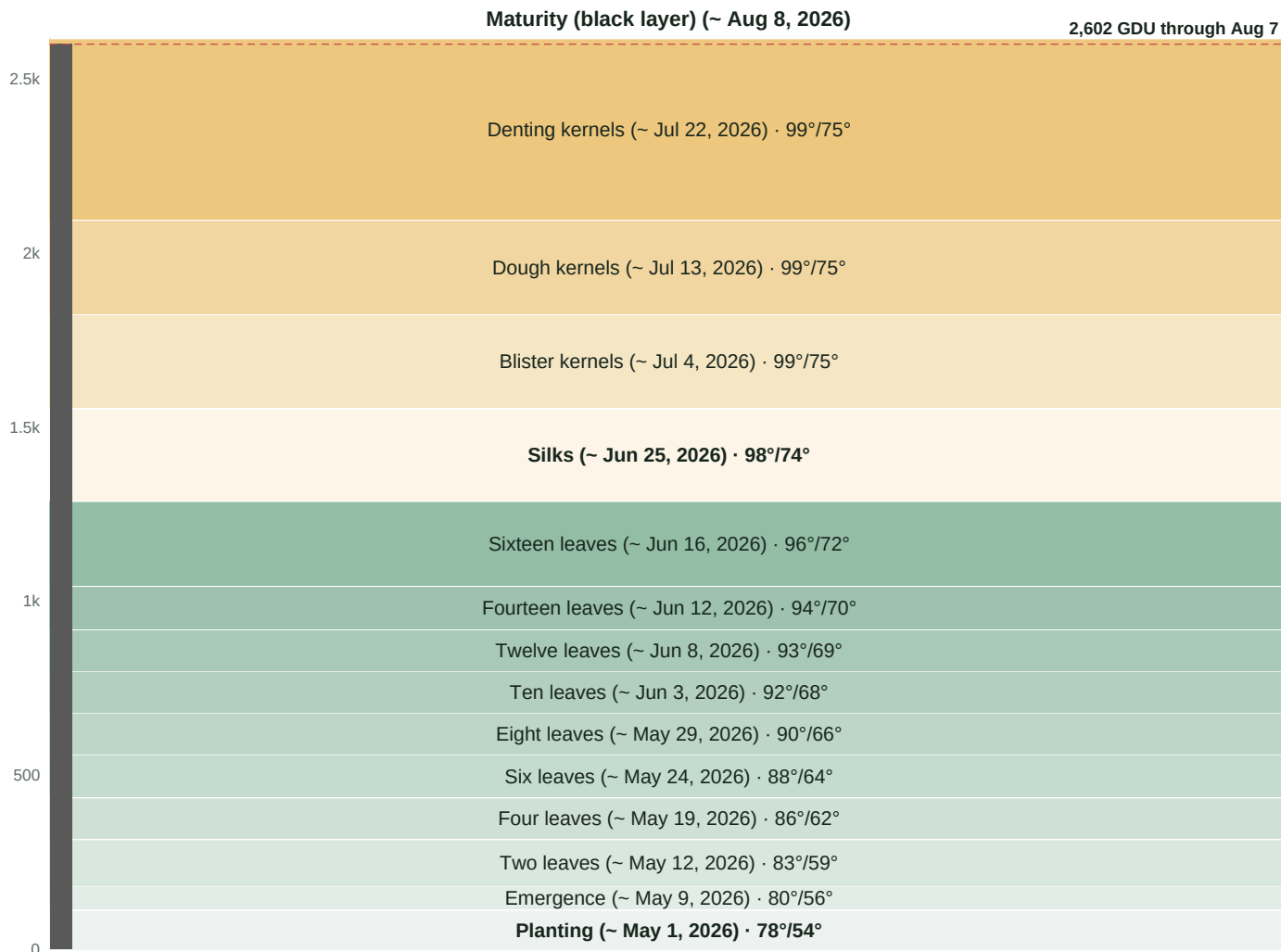
Predicted Stage Dates

SCENARIO	SILK (1,290)	BLACK LAYER (2,620)
This season (2026)	Jun 25, 2026 (reached)	Aug 8, 2026 (in forecast)
Last year (2025) actual	Jun 28, 2026	Aug 13, 2026
Abnormally hot year (90th pct)	Jun 22, 2026	Aug 4, 2026
Normal (30-yr median)	Jun 30, 2026	Aug 16, 2026
Abnormally cool year (10th pct)	Jul 8, 2026	Aug 29, 2026

"Reached" is read off observed weather and cannot move. "In forecast" is inside the 16-day outlook. Only a "projected" date carries a hot-to-cool range, and that range is the answer, not the single date above it. The rows beneath are whole seasons for comparison.

Growth Stages

Dates shown for: This season — normal finish. Planting, Silks and Maturity sit exactly on the hybrid's own numbers; every stage between them is scaled from Iowa State's published GDU ladder and is an estimate.



Frost Risk

Oct 29

28 °F BY THIS DATE 1 YR IN 10

Nov 7

MEDIAN 28 °F FREEZE

Oct 30

MEDIAN 32 °F FROST

82 days of margin even against a 1-year-in-10 early freeze (Oct 29), 91 against the median. Comfortable for this location and planting date.

Read these as the LATE end of the range: a 6-to-15 mile grid cell averages away the radiative cooling that makes a low spot frost first, so against nearby station records these dates run 1 to 3 weeks late. GDU accumulation itself is within about 1% of station records.

How These Numbers Were Made

- GDU = $(\min(\text{daily high}, 86^\circ\text{F}) + \max(\text{daily low}, 50^\circ\text{F})) \div 2 - 50$ — the modified base-50/86 method US seed companies rate hybrids on. A day below 50 °F counts 0, never a negative; heat above 86 °F adds nothing. Accumulation starts on the planting date itself.
- Normal, hot and cool are the 50th, 90th and 10th percentiles of accumulation across 30 complete years (1996–2025) at this exact grid point — an envelope, not a replay of any one year. Growth stages between Planting, Silks and Maturity are scaled from Iowa State's published ladder for a 2,700-GDU hybrid and are estimates.
- Temperatures: ERA5 reanalysis via Open-Meteo for history and the current season through Aug 7, 2026, plus Open-Meteo's 16-day forecast through Aug 22. Grid point 41.5644, -95.8913.
- GDU is a heat model, not a crop model. It knows nothing about drought, saturated soils, replant, hail, disease or nitrogen — any of which can move real silk and black layer dates well off these numbers.